CLASS 10 · CBSE · NCERT MATHEMATICS · NCERT SOLUTIONS

Chapter 11: Areas Related to Circles

Step-by-step solutions to the textbook exercises: 14 questions

Questions are numbered as in the textbook. Try each one yourself before reading the solution.

1 Exercise 11.1

Q1 Find the area of a sector of a circle of radius 6 cm with sector angle 60°. (Use $\pi = \frac{22}{7}$ unless told otherwise.)

SOLUTION

Area of a sector $= \frac{\theta}{360} \times \pi r^2 = \frac{60}{360} \times \frac{22}{7} \times 6 \times 6 = \frac{132}{7}$.

Answer: $\frac{132}{7}$ **cm^2 (about 18.86 cm^2)**

Q2 A circle has a circumference of 22 cm. What is the area of one quadrant of it?

SOLUTION

$2\pi r = 22$ gives $r = \frac{22 \times 7}{2 \times 22} = \frac{7}{2}$ cm.

Area of a quadrant $= \frac{1}{4}\pi r^2 = \frac{1}{4} \times \frac{22}{7} \times \frac{7}{2} \times \frac{7}{2} = \frac{77}{8}$.

Answer: $\frac{77}{8}$ **cm^2**

Q3 The minute hand of a clock is 14 cm long. Find the area it sweeps in 5 minutes.

SOLUTION

In 60 minutes the hand turns through 360°, so in 5 minutes it turns through 30°.

Area $= \frac{30}{360} \times \frac{22}{7} \times 14 \times 14 = \frac{154}{3}$.

Answer: $\frac{154}{3}$ **cm^2 (about 51.33 cm^2)**

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Q4 In a circle of radius 10 cm, a chord makes a right angle at the centre. Calculate (i) the area of the minor segment cut off by the chord (ii) the area of the major sector. (Use $\pi = 3.14$.)

SOLUTION

Area of the minor sector $= \frac{90}{360} \times 3.14 \times 100 = 78.5$ cm^2. Area of the right triangle formed by the radii $= \frac{1}{2} \times 10 \times 10 = 50$ cm^2.

(i) Minor segment $= 78.5 - 50 = 28.5$ cm^2.

(ii) Major sector $=$ area of the circle $-$ minor sector $= 314 - 78.5 = 235.5$ cm^2.

Answer: (i) 28.5 cm^2 (ii) 235.5 cm^2

Q5 An arc of a circle of radius 21 cm makes an angle of 60° at the centre. Find (i) the length of the arc (ii) the area of the sector (iii) the area of the segment cut off by the chord joining the ends of the arc.

SOLUTION

(i) Arc length $= \frac{60}{360} \times 2 \times \frac{22}{7} \times 21 = 22$ cm.

(ii) Area of the sector $= \frac{60}{360} \times \frac{22}{7} \times 21 \times 21 = 231$ cm^2.

(iii) The triangle formed by the two radii and the chord is equilateral (two equal sides with an angle of 60° between them). Its area is $\frac{\sqrt{3}}{4} \times 21^2 = \frac{441\sqrt{3}}{4}$ cm^2. Segment $= 231 - \frac{441\sqrt{3}}{4}$ cm^2.

Answer: (i) 22 cm (ii) 231 cm^2 (iii) $\left(231 - \frac{441\sqrt{3}}{4}\right)$ cm^2

Q6 A chord of a circle of radius 15 cm makes an angle of 60° at the centre. Calculate the areas of the minor segment and of the major segment. (Use $\pi = 3.14$ and $\sqrt{3} = 1.73$.)

SOLUTION

Area of the sector $= \frac{60}{360} \times 3.14 \times 225 = 117.75$ cm^2.

The triangle is equilateral with side 15 cm: area $= \frac{\sqrt{3}}{4} \times 225 = 1.73 \times 56.25 = 97.3125$ cm^2.

Minor segment $= 117.75 - 97.3125 = 20.4375$ cm^2. Major segment $= 3.14 \times 225 - 20.4375 = 706.5 - 20.4375 = 686.0625$ cm^2.

Answer: minor segment 20.4375 cm^2, major segment 686.0625 cm^2

Q7 In a circle of radius 12 cm, a chord makes an angle of 120° at the centre. Find the area of the minor segment it cuts off. (Use $\pi = 3.14$ and $\sqrt{3} = 1.73$.)

SOLUTION

Area of the sector $= \dfrac{120}{360} \times 3.14 \times 144 = 150.72$ cm^2.

Draw the perpendicular from the centre to the chord. It makes two right triangles with angles of 60° at the centre: the height is $12\cos 60^\circ = 6$ cm and half the chord is $12\sin 60^\circ = 6\sqrt{3}$ cm.

Area of the triangle $= \dfrac{1}{2} \times 12\sqrt{3} \times 6 = 36\sqrt{3} = 62.28$ cm^2.

Answer: segment $= 150.72 - 62.28 = 88.44$ cm^2

Q8 A horse is tied to a peg at one corner of a square field of side 15 m by a rope 5 m long. Find (i) the area of the field the horse can graze (ii) the increase in the grazing area if the rope were 10 m long. (Use $\pi = 3.14$.)

SOLUTION

The corner angle is 90°, so the horse grazes a quadrant of a circle whose radius is the length of the rope.

(i) $\dfrac{1}{4} \times 3.14 \times 5^2 = 19.625$ m^2.

(ii) With a 10 m rope: $\dfrac{1}{4} \times 3.14 \times 10^2 = 78.5$ m^2. Increase $= 78.5 - 19.625 = 58.875$ m^2.

Answer: (i) 19.625 m^2 (ii) 58.875 m^2

Q9 A brooch is made of silver wire as a circle of diameter 35 mm, with 5 diameters that divide it into 10 equal sectors. Find (i) the total length of wire needed (ii) the area of each sector.

SOLUTION

(i) Circumference $= \dfrac{22}{7} \times 35 = 110$ mm. Five diameters $= 5 \times 35 = 175$ mm. Total $= 285$ mm.

(ii) Each sector has angle 36°: area $= \dfrac{1}{10} \times \dfrac{22}{7} \times \dfrac{35}{2} \times \dfrac{35}{2} = \dfrac{385}{4}$ mm^2.

Answer: (i) 285 mm (ii) $\dfrac{385}{4}$ mm^2 (96.25 mm^2)

Q10 An umbrella has 8 equally spaced ribs. Taking it as a flat circle of radius 45 cm, find the area between two consecutive ribs.

SOLUTION

The ribs divide the circle into 8 equal sectors, each with angle 45°.

Area $= \dfrac{1}{8} \times \dfrac{22}{7} \times 45 \times 45 = \dfrac{22275}{28}$.

Answer: $\dfrac{22275}{28}$ cm^2 (about 795.54 cm^2)

Q11 A car has two wipers that do not overlap. Each blade is 25 cm long and sweeps through 115°. Find the total area cleaned at each sweep.

SOLUTION

Area swept by one blade $= \frac{115}{360} \times \frac{22}{7} \times 25 \times 25$.

Total for two blades $= 2 \times \frac{115}{360} \times \frac{22}{7} \times 625 = \frac{158125}{126}$.

Answer: $\frac{158125}{126}$ cm^2 (about 1254.96 cm^2)

Q12 A lighthouse sends out a red beam that covers a sector of 80° and reaches 16.5 km. How much sea area is covered by the warning beam? (Use $\pi = 3.14$.)

SOLUTION

Area $= \frac{80}{360} \times 3.14 \times 16.5 \times 16.5 = \frac{2}{9} \times 3.14 \times 272.25 = 189.97$.

Answer: about 189.97 km^2

Q13 A round table cover of radius 28 cm has six equal designs, each the segment between a side of a regular hexagon drawn in the circle and the arc above it. Find the cost of making the designs at ₹0.35 per cm^2. (Use $\sqrt{3} = 1.7$.)

SOLUTION

Each side of the hexagon subtends 60° at the centre, so each design is a segment of angle 60°, and its triangle is equilateral with side 28 cm.

One sector $= \frac{60}{360} \times \frac{22}{7} \times 28 \times 28 = \frac{1232}{3}$ cm^2. One triangle $= \frac{\sqrt{3}}{4} \times 28^2 = 1.7 \times 196 = 333.2$ cm^2.

Six designs $= 6\left(\frac{1232}{3} - 333.2\right) = 2464 - 1999.2 = 464.8$ cm^2.

Cost $= 464.8 \times 0.35 =$ ₹162.68.

Answer: ₹162.68

Q14 The area of a sector of angle p (in degrees) of a circle of radius R is (A) $\frac{p}{180} \times 2\pi R$ (B) $\frac{p}{180} \times \pi R^2$ (C) $\frac{p}{360} \times 2\pi R$ (D) $\frac{p}{720} \times 2\pi R^2$.

SOLUTION

Area of a sector $= \frac{p}{360} \times \pi R^2 = \frac{p}{720} \times 2\pi R^2$.

Answer: (D)

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